

From $\mathbb{Z}_3$ to Three Generations: A Structural Bridge between PDL Leakage Cycles and OFN Fermion Families

Joint PDL–OFN Note N02

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Status of this document

This document is a joint draft of N02, integrating the complete PDL-side (Sections 1–3, 5–6) and the complete OFN-side (Section 4). The comparison table (Section 3) and open problems (Section 7) have been updated to reflect all results established in July 2026. The document is ready for final joint review before Zenodo deposit.

Abstract

The joint note N01 [1] established that the first Betti number $\beta_1(K_4) = 3$ (Projective Dynamic Logo, PDL) and the cyclomatic number $b_1(\Omega_{21}) = 3$ (Ontological Fundamental Network, OFN) are the same topological invariant arising independently in two unrelated combinatorial frameworks. Subsequent PDL documents (D57–D59) trace this invariant to a specific algebraic origin internal to PDL: the quotient $A_4/V_4 \cong \mathbb{Z}_3$ in $\text{Sym}(K_4) = S_4$, which is simultaneously the centre of the colour gauge group $\text{SU}(3)$ (D58) and the index set of the three axes of the fundamental representation **3** (D59). The present note examines whether this $\mathbb{Z}_3$ structure, established as an unconditional PDL theorem, can be placed in correspondence with the three fermion generations of the OFN framework, thereby providing a partial resolution of the open problem OP-OFN-1. We present the complete PDL-side derivation chain together with a candidate structural analogy to a numerically verified arithmetic identity linking the PDL hadronic and cosmological sectors ($n_u - 1 = p_{k_1} = 23$), and we propose a comparison table format, following the working conditions of N01, within which the OFN-side correspondence — including the structural self-reference operator S_{sr} — is to be developed jointly.

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1 Introduction and motivation

N01 [1] posed three open problems at the interface of the PDL and OFN frameworks. Of these, OP-OFN-1 — the formal link between the three PDL leakage cycles and the three OFN fermion generations — has remained the most resistant, because at the time of N01 neither framework had yet identified the precise algebraic mechanism generating its respective invariant 3.

Since N01, three PDL documents have closed part of this gap from the PDL side. D57 [2] identified the quotient $A_4/V_4 \cong \mathbb{Z}_3$ within $\text{Sym}(K_4) = S_4$ as a candidate combinatorial origin of the invariant. D58 [3] proved, as an unconditional algebraic theorem of the PDL axioms C1–C4, that this same quotient $\mathbb{Z}_3$ is exactly the centre of the colour gauge group $\text{SU}(3)$. D59 [4] went further, constructing explicitly the fundamental representation **3** of $\text{SU}(3)$ on the trace-zero plane $W \subset \mathbb{C}^3$, whose three axes are canonically labelled both by $V_4 \setminus \{e\}$ (the geometric labelling) and by the triplet $(T_2) = \{k_1, k_2, k_3\}$ of leakage-cycle indices from D51–D52 (the dynamical labelling).

The purpose of the present note is twofold. First, to assemble this PDL-side chain in a single self-contained account, so that the precise algebraic content of the invariant 3 is available for comparison with its OFN counterpart. Second, to propose — under the same epistemic discipline as N01 — a structured format within which the OFN-side mechanism (in particular the structural self-reference operator S_{sr} introduced in the OFN programme [5]) can be set alongside the PDL result without conflating the two frameworks.

A secondary, more speculative thread is introduced in Section 6: the arithmetic identity $n_u - 1 = p_{k_1} = 23$ (D63), which connects the PDL hadronic sector (the proton valence multiplicity n_u) to the PDL cosmological sector (the first leakage-cycle prime p_{k_1}). This identity is independent of, and additional to, the main $\mathbb{Z}_3$ chain; it is presented separately and at a lower epistemic register, consistent with its open status (OP-D63-3) in the PDL corpus.

2 The PDL side: from $A_4/V_4 \cong \mathbb{Z}_3$ to the three axes of W

2.1 The elementary closure and its automorphisms

The PDL axioms C1–C4 force the elementary closure to be the complete signed graph K_4 on four vertices and six edges [6]. The automorphism group of K_4 as an unsigned graph is S_4 , acting on the vertex set by permutation. Two distinguished normal subgroups of S_4 are central to the chain assembled here: the Klein four-group $V_4 = \{e, (01)(23), (02)(13), (03)(12)\}$, and the alternating subgroup A_4 of even permutations, with the inclusion $V_4 \subset A_4 \subset S_4$ and indices $[A_4 : V_4] = 3$, $[S_4 : A_4] = 2$.

2.2 Three theorems: S_3 , $\mathbb{Z}_3$, and the representation 3

PDL (established)

Theorem (D60). The physical indistinguishability of the two states of a pulsation pairing, forced by axioms C1 and C2 alone, implies that every C1-admissible observable on $\text{Coh}(K_4)$ is invariant under V_4 . Consequently the effective symmetry group of K_4 is

$$G_{\text{eff}} = S_4/V_4 \cong S_3.$$

This is an unconditional theorem of C1 and C2, with no auxiliary hypothesis.

PDL (established)

Theorem (D58, Lemma L3). The quotient A_4/V_4 is cyclic of order 3:

$$A_4/V_4 \cong \mathbb{Z}_3,$$

and under the homomorphism $\rho: S_4 \rightarrow S_3$ of the previous theorem, the image of A_4 is exactly the alternating subgroup $\text{Alt}(3) \subset S_3$ — the centre-generating cyclic subgroup. D58 further proves, via the Cartan–Killing–Weyl correspondence and the A_2 root system realised in the rank-2 reduction forced by $R_e = 6$, that this $\mathbb{Z}_3$ is exactly the centre $Z(\text{SU}(3))$ of the colour gauge group. The derivation of $\text{SU}(3)$ as an algebraic theorem of C1–C4 is complete.

PDL (established)

Theorem (D59). The fundamental representation $\mathbf{3}$ of $\text{SU}(3)$ is realised on the trace-zero plane $W = \{(a, b, c) \in \mathbb{C}^3 : a + b + c = 0\}$. The three axes of W are canonically labelled, in an S_4 -equivariant manner, by the three non-identity elements of V_4 . The same three axes admit a second, dynamically motivated labelling by the triplet

$$(T_2) = \{k_1, k_2, k_3\} = \{9, 19, 168\},$$

the indices of the three independent leakage cycles of D51–D52 associated with the cosmological constant Λ_{PDL} . The matter orientation $\mathbf{3}$ (as opposed to $\bar{\mathbf{3}}$) is forced, without additional input, by the isospin asymmetry $\Delta n = n_d - n_u = 4 > 0$, itself an unconditional theorem of C4.

Remark 2.1 (What is and is not established). The chain above establishes, as unconditional theorems of C1–C4, that: (i) the invariant 3 of N01 originates algebraically in the quotient $A_4/V_4 \cong \mathbb{Z}_3$; (ii) this $\mathbb{Z}_3$ is simultaneously the centre of the colour gauge group $\text{SU}(3)$ and the index set of the fundamental representation’s three axes; (iii) the same three axes carry a second labelling by the three leakage-cycle indices (k_1, k_2, k_3) of the cosmological sector. What is *not* established by this chain, and remains the subject of the present note, is any connection between this $\mathbb{Z}_3$ structure and the three *fermion generations* of either framework. The PDL corpus itself states this explicitly as OP-D59-1 (quark mass spectrum and generation structure from C1–C4 is open) and OP-D61-2 (three fermion generations from C1–C4 is open).

3 The PDL leakage cycles: dynamical content of (T_2)

The triplet $(T_2) = \{k_1, k_2, k_3\}$ is not merely a relabelling of $V_4 \setminus \{e\}$; it carries independent dynamical content established in D51–D52, which is the natural point of contact with the OFN generation structure.

PDL (established)

Theorem (D51, Lemma 1). The first Betti number of the 1-skeleton of K_4 is $\beta_1(K_4) = |E| - |V| + 1 = 6 - 4 + 1 = 3$: a purely topological fact, independent of sign assignment or dynamics. There are exactly three independent leakage cycles.

PDL (established)

Theorem (D51, Lemma 2–3; D52). Axioms C1 and C3 force the lengths of admissible leakage cycles to be prime. The three leakage cycles of K_4 have prime exponents

$$(k_1, k_2, k_3) \mapsto (p_{k_1}, p_{k_2}, p_{k_3}) = (23, 67, 997),$$

and the corresponding leakage bases are uniquely identified (D52) as $(1 - \kappa)$, $r_{\text{val}}/R_{\text{tot}}$, and $(1 - \eta_L)$ respectively, each itself an unconditional theorem of C1–C4. The resulting cosmological leakage constant C reproduces Λ_{obs} to 0.17 parts per million (Planck 2020).

Remark 3.1. It is this triplet of *prime-indexed, dynamically active* leakage cycles — not merely the abstract group $\mathbb{Z}_3$ — that is the most natural PDL object to compare against three OFN fermion generations, since fermion generations in any framework are expected to carry distinguishing dynamical content (e.g. mass hierarchy), not merely a permutation symmetry.

4 Comparison table: PDL result | OFN result | status of identification

Following the working conditions agreed for this note (Section 7), every cross-framework correspondence is recorded in the table below rather than in a shared equation. The PDL column is complete; the OFN column and the status column are to be completed jointly.

Table 1: Comparison table for the $\mathbb{Z}_3$ / three-generations correspondence and the five Session 67 connections. Status labels follow the PDL epistemic convention: **mathematical identity**, **structural echo** / **candidate analogy**, **open problem**.

PDL result	OFN result	Status
$\beta_1(K_4) = 3$ (topological, D51 Lemma 1)	$b_1(\Omega_{21}) = 3$ AND $b_1(G_H) = 3$ (N01 + Bachani CWS code, Sec. 5.1)	Mathematical identity (N01): same invariant, independent derivations.
$A_4/V_4 \cong \mathbb{Z}_3 \subset \text{Sym}(K_4)$ (D58 Lemma L3)	$b_1(G_H) = 3$ cycles; the tripartition $\{A, B, C\}$ of $K_{2,2,2} = L(K_4)$ is the set of three perfect matchings of K_4 (Sec. 5.7)	Mathematical identity: both are the same three-element set (matchings of $K_4 = V_4 \setminus \{e\}$).
$\mathbb{Z}_3 = Z(\text{SU}(3))$, centre of colour gauge group (D58)	$13 = 8 \oplus 3 \oplus 1 \oplus 1$ (CWS code decomp.) encodes gauge group; $\mathbb{Z}_3$ is the centre of $\text{SU}(3)_c$	Shares integer 3; no structural identification claimed beyond this. OP-OFN-3 open.
Three axes of W , labelled by $V_4 \setminus \{e\}$ and by $(T_2) = \{k_1, k_2, k_3\}$ (D59)	Three cycles of G_H with $b_1(G_H) = 3$; S_3 broken to Z_2 by holonomy distinction $\Phi_1 = \Phi_2 = 2\pi/12$, $\Phi_3 = 5\pi/12$	OP-OFN-1 partial resolution: two classes distinguished by holonomy; full three-generation separation open (OP-N02-2).
ϕ in $\kappa = 310\phi/11017$ (D44, unconditional theorem)	$\gamma_E = 3 - \sqrt{5} = 4 - 2\phi$, spectral gap of G_E (dodecahedron on Ω_{21} , Theorem 5.6)	Algebraic identity: $\phi + \gamma_E/2 = 2$ exact in $\mathbb{Q}(\sqrt{5})$ (Theorem 5.7). Note: G_H has a different gap $\lambda_1 \approx 0.0804$, algebraically unrelated to ϕ .
4 vertices +6 edges of $K_4 = 10$ graph elements (D16a)	$\tau = 90 = b_1^2 \times \dim P(1,3) = 9 \times 10$ (Corollary 5.10); $\dim P(1,3) = 4 + 6 = 10$	Structural echo: OP-OFN-2 open.
$(p_{k_1}, p_{k_2}, p_{k_3}) = (23, 67, 997)$, prime exponents of three leakage cycles (D51–D52)	Three CP-pairs + 13 unpaired states (Theorem 5.11); no OFN-side mass hierarchy from leakage primes yet identified	OP-OFN-1 and OP-N02-1 open.
$n_u - 1 = p_{k_1} = 23$ (D47+D51; isolation 0.265% / 378 pairs)	No direct OFN counterpart identified at present	Speculative extension; OP-D63-3 open.

5 The OFN side: Spectral Structure of the Vacuum Manifold Ω_{21}

This section presents the OFN-side results that complement the PDL-side of this bridge note. All results concern the vacuum manifold Ω_{21} , a 21-vertex subset of the 6-dimensional Boolean hypercube Q_6 , and its spectral decomposition under the graph Laplacian.

5.1 Definition of the Vacuum Manifold

The vacuum manifold is defined as the unique 21-state subset of Q_6 selected by three cascading self-consistency conditions [7]:

1. **Causal regularity:** vertex-transitive cubic subgraphs of Q_6 at Hamming distance 2.

2. **Cage classification:** retention of graphs with girth ≥ 5 (by Tutte’s cage theorem, only the Petersen graph $P(10)$ and the dodecahedron $D(20)$ qualify).
3. **Binary embedding:** self-consistent embedding $\Phi^*: \text{graph} \rightarrow Q_6$ preserving adjacency at Hamming distance 2. Only $D(20)$ admits such an embedding.

The Hamming-isolated state $\sigma = 21 = (010101)_2$ completes the vacuum:

$$\Omega_{21} = \Phi^*(D_{20}) \cup \{\sigma\} = \{0, 1, 3, 4, 7, 8, 9, 12, 15, 16, 19, 21, 27, 31, 35, 42, 43, 48, 52, 56, 63\}$$

5.2 Two Distinct Graphs on Ω_{21}

Critical clarification. The vacuum manifold Ω_{21} carries *two distinct graphs*, each with its own spectral gap. This distinction was not explicit in earlier OFN literature and is essential for the PDL–OFN bridge.

Definition 5.1 (Graph G_E — Dodecahedron). The graph G_E is the dodecahedron $D(20)$, with vertex set $M_{20} = \Omega_{21} \setminus \{\sigma\}$ (the 20-vertex connected component) and edges defined by the graph structure of $D(20)$ embedded in Q_6 via the map Φ^* [7]. The graph G_E is 3-regular with 30 edges. The state $\sigma = 21$ is *not* part of G_E ; it is an isolated vertex (degree 0) in the full graph on Ω_{21} .

Definition 5.2 (Graph G_H — Induced Subgraph). The graph G_H has both endpoints in $M_{20} = \Omega_{21} \setminus \{\sigma\}$, with edges at Hamming distance 1. It has 22 edges and is irregular, with vertex degrees $\{1, 2, 3, 4\}$.

Physical interpretation. The two graphs correspond to different physical roles [7]: G_E governs kinetic energy and propagation (dodecahedral topology); G_H encodes internal topology and the three Betti cycles.

5.3 Spectral Decomposition of G_H

Theorem 5.3 (Spectral Factorization over $\mathbb{Q}$). *The characteristic polynomial of the Laplacian of the 20-vertex connected component of G_H factorizes exactly over the rationals as:*

$$P_{20}(x) = x \cdot (x - 1)^2 \cdot (x - 3) \cdot P_{16}(x)$$

where $P_{16}(x)$ is an irreducible polynomial of degree 16 over $\mathbb{Q}$ with constant term 600.

Proof. The factorization was verified computationally by exact integer arithmetic (Python, NumPy, NetworkX) and confirmed independently by both authors (July 2026). The irreducibility of $P_{16}(x)$ was verified by checking that it has no rational roots and does not factor over $\mathbb{Q}$ into polynomials of lower degree. $\square$

Corollary 5.4 (Exact Integer Eigenvalues). *The Laplacian of G_H has exact integer eigenvalues: $\lambda = 0$ (multiplicity 2, one per connected component), $\lambda = 1$ (multiplicity 2), and $\lambda = 3$ (multiplicity 1). The remaining 16 eigenvalues are algebraic numbers of degree 16, roots of $P_{16}(x)$.*

Theorem 5.5 (Spectral Gap of G_H). *The spectral gap of G_H (smallest nonzero eigenvalue of the Laplacian) is:*

$$\lambda_1 \approx 0.0804170036$$

This is the smallest root of $P_{16}(x)$ and is an algebraic number of degree 16.

This value was computed independently by both authors, agreeing to 10 significant figures.

5.4 Spectral Gap of G_E and the Algebraic Identity

Theorem 5.6 (Spectral Gap of G_E). *The spectral gap of G_E (the dodecahedron) is $\gamma_E = 3 - \sqrt{5} \approx 0.7639$, a known mathematical fact for the dodecahedron graph.*

Theorem 5.7 (Algebraic Identity in $\mathbb{Q}(\sqrt{5})$). *The following identity holds exactly in $\mathbb{Q}(\sqrt{5})$:*

$$\phi + \frac{\gamma_E}{2} = 2$$

where $\phi = (1 + \sqrt{5})/2$ is the golden ratio and $\gamma_E = 3 - \sqrt{5}$ is the spectral gap of G_E .

Proof. Direct computation: $\phi + \frac{\gamma_E}{2} = \frac{1 + \sqrt{5}}{2} + \frac{3 - \sqrt{5}}{2} = \frac{4}{2} = 2$. $\square$

Corollary 5.8 (Non-Identity for G_H). *The analogous identity does not hold for G_H : $\phi + \lambda_1/2 \approx 1.658 \neq 2$. The algebraic identity $\phi + \gamma/2 = 2$ is specific to the dodecahedron G_E , not the induced subgraph G_H .*

5.5 The Spanning Tree Invariant $\tau = 90$

Theorem 5.9 (Spanning Trees of G_H). *The number of spanning trees of the 20-vertex connected component of G_H is $\tau = 90$.*

Proof. By the Matrix Tree Theorem (Kirchhoff), $\tau = \frac{1}{n} \prod_{i=1}^{n-1} \lambda_i$ where $n = 20$. From Theorem 5.3: $\prod_{i=1}^{19} \lambda_i = 1^2 \cdot 3 \cdot 600 = 1800$, so $\tau = 1800/20 = 90$. $\square$

This was verified numerically to machine precision (product of nonzero eigenvalues = 1800.0000) by both authors.

Corollary 5.10 (Structural Echo). *The invariant $\tau = 90$ admits the factorization:*

$$\tau = 90 = 9 \times 10 = b_1^2 \times \dim P(1, 3)$$

where $b_1 = 3$ is the first Betti number of G_H and $\dim P(1, 3) = 10$ is the dimension of the Poincaré group (4 translations + 6 Lorentz generators). **Epistemic status:** this is a structural echo, not a proven identity (see OP-OFN-2 below).

5.6 CP-Pair Classification

Theorem 5.11 (CP-Pair Classification). *Under the CP involution $s \mapsto 63 - s$, the 21 states of Ω_{21} decompose into exactly 4 CP-pairs (8 states) and 13 unpaired states. The 4 CP-pairs are: (0, 63), (7, 56), (15, 48), (21, 42).*

Proof. The CP involution has no fixed points in $\mathbb{Z}$ (since $s = 63 - s$ implies $s = 31.5$). Enumeration of all pairs $(s, 63 - s)$ with both elements in Ω_{21} yields exactly the four pairs listed. The remaining 13 states have their CP-partner outside Ω_{21} . Consistency: $4 \times 2 + 13 = 21$. $\square$

Remark 5.12. The state $\sigma = 21 = (010101)_2$ is Hamming-isolated in Ω_{21} (degree 0 in G_H), but is *not* a fixed point of the CP involution: its CP-partner is $42 = (101010)_2 \in \Omega_{21}$, forming the fourth CP-pair. This classification agrees with Bachani [7], who identifies the 8 states of the 4 CP-pairs as “matter states” and the 13 unpaired states as “gauge channels” — where “self-conjugate” in Bachani’s terminology refers to membership in a CP-pair within Ω_{21} , not to fixed points of the involution.

5.7 Summary of the Five Session 67 Connections

The following five connections between PDL and OFN were established in July 2026:

1. **Mathematical identity (tripartition):** The tripartition $\{A, B, C\}$ of $K_{2,2,2} = L(K_4)$ is exactly the set of three perfect matchings of K_4 (PDL D58 Lemma L2, D61). *Status: mathematical identity.*
2. **Algebraic identity in $\mathbb{Q}(\sqrt{5})$:** $\phi + \gamma_E/2 = 2$ (Theorem 5.7). *Status: algebraic identity.*
3. **Corrected automorphism group:** $\text{Aut}(K_{2,2,2}) = S_2 \wr S_3$, order 48 (O_h), not S_4 ; Whitney's theorem does not apply to K_4 . *Status: corrected computation.*
4. **Dimensional echo:** K_4 has 4 vertices and 6 edges, $4 + 6 = 10 = \dim P(1, 3)$. *Status: structural echo.*
5. **Hadronic-cosmological identity:** $n_u - 1 = p_{k_1} = 23$ (D47 + D51), verified 13/13 at isolation 0.265% against 378 pre-registered pairs. *Status: verified identity.*

5.8 Epistemic Summary and Verification Record

Unconditional theorems: Spectral factorization $P_{20}(x)$ (Theorem 5.3); spectral gap $\lambda_1 \approx 0.0804$ of G_H (Theorem 5.5); spectral gap $\gamma_E = 3 - \sqrt{5}$ of G_E (Theorem 5.6); algebraic identity $\phi + \gamma_E/2 = 2$ (Theorem 5.7); spanning tree invariant $\tau = 90$ (Theorem 5.9); CP-pair classification (Theorem 5.11).

Structural echoes (not proven identities): $\tau = 90 = b_1^2 \times \dim P(1, 3)$; $4 + 6 = 10 = \dim P(1, 3)$.

All computations were performed by exact integer arithmetic (Python, NumPy, NetworkX) and independently verified by both authors in July 2026. Scripts are available upon request.

6 Speculative extension: the identity $n_u - 1 = p_{k_1} = 23$

Speculative extension

This section is presented at a lower epistemic register than Sections 2–3. The chain $A_4/V_4 \cong \mathbb{Z}_3 \rightarrow (T_2) \rightarrow \text{axes of } W$ is a sequence of unconditional theorems. The material below is a numerically verified *coincidence* whose structural origin is an open problem (OP-D63-3) of the PDL corpus. It is included here because it independently connects the same leakage index k_1 to a second PDL sector (hadronic masses), and may therefore be of interest as an additional, lower-confidence data point for the OFN comparison — not as a second derivation chain of comparable strength to Sections 2–3.

6.1 The identity

D63 [8] records the following exact integer identity between two quantities derived from independent PDL mechanisms:

$$n_u - 1 = p_{k_1} = 23, \tag{1}$$

where $n_u = 24$ is the up-quark valence multiplicity of the proton quintuplet (D47, forced by the quasi-completeness constraint $3n_u^2 + 5n_u - 1848 = 0$, discriminant 149^2), and $p_{k_1} = 23$ is the first leakage-cycle prime of the cosmological sector (D51, with $k_1 = n_d - n_u + R_e - 1 = 9$ and p_{k_1} the 9th prime).

6.2 Verification and isolation test

The identity (1) and its statistical isolation within the natural reference class of simple linear expressions in the quintuplet integers were verified independently in the script `PDL_N02_identity_lockdown_v2_reinforced.py` [9]. The script: (i) recomputes $n_u = 24$ from the quasi-completeness constraint independently of its statement in D63; (ii) recomputes $k_1 = 9$ and $p_{k_1} = 23$ from the leakage-index definition of D51; (iii) verifies the identity (1) by exact integer arithmetic; (iv) tests a pre-registered family of 378 candidate (index, value) expression pairs built from simple linear combinations of $(n_u, n_d, R_e, \Delta n)$, finding that only 1 pair — the documented identity itself — reproduces an exact prime-index match (isolation fraction 0.265%); (v) cross-checks this rate against a randomised null model over the same coefficient ranges (5000 draws, match rate 0.059%), confirming the pre-registered family was not tuned to produce an artificially low rate. All checks passed (13/13).

Remark 6.1 (Epistemic status). This result supports treating the identity (1) as a non-trivial structural coincidence — it is not a generic feature of prime density, nor an artefact of a cherry-picked comparison family — but it does *not* constitute a derivation from C1–C4. The structural origin of the identity remains OP-D63-3, open in the PDL corpus. No claim is made here regarding any OFN-side counterpart of this specific identity; this section is offered to Oleg for consideration only.

7 Working conditions for this note

Consistent with the conditions agreed by both authors prior to drafting, and modelled on the epistemic discipline of N01:

1. Every correspondence between a PDL object and an OFN object is recorded as a *candidate structural analogy* (Table 1) until a derivation common to both frameworks exists.
2. The structural self-reference operator S_{sr} is introduced strictly as an OFN-side concept originating in [5], clearly attributed, and is not identified with any PDL axiom or theorem.
3. No equation in this note mixes objects native to both frameworks; cross-framework statements are confined to Table 1 and to remarks explicitly labelled as candidate analogies.
4. Results are stratified by epistemic status throughout: *unconditional theorem*, *corroborated conjecture*, *candidate structural analogy*, or open problem, following the convention of the PDL corpus [10].

8 Open problems

Open Problem 8.1 (OP-OFN-1 — restated). Formally identify the $\mathbb{Z}_3$ structure of Section 2 with the three fermion generations of OFN. The holonomy mechanism of Section 5 (brisure $S_3 \rightarrow Z_2$, $\Phi_1 = \Phi_2 \neq \Phi_3$) provides a partial resolution: two classes are distinguished dynamically. The full separation of all three generations and the mass hierarchy remain open.

Open Problem 8.2 (OP-D63-3 — restated). Determine whether the identity $n_u - 1 = p_{k_1} = 23$ (Section 6) is forced by C1–C4 or is a coincidence at the current level of approximation. **Entry point:** D47, D51, D63, present note.

Open Problem 8.3 (New — OP-N02-1). Determine whether the dynamical content of the three leakage cycles (prime exponents 23, 67, 997, Section 3) admits any OFN-side counterpart distinguishing the three fermion generations dynamically (e.g. a mass-hierarchy mechanism), beyond the permutation symmetry $\mathbb{Z}_3$ common to both frameworks. **Entry point:** D51, D52, G_H holonomy (Section 5).

Open Problem 8.4 (New — OP-N02-2). Break the residual Z_2 symmetry between cycles C_1 and C_2 of G_H (which share equal holonomies $\Phi_1 = \Phi_2 = 2\pi/12$, a topological consequence of their identical edge-class composition). This requires a more refined phase rule Θ accounting for additional structure (e.g. Hamming weights of specific vertices or higher-order holonomy corrections). **Entry point:** Section 5, $G_H(\Omega_{21})$.

Open Problem 8.5 (New — OP-OFN-2). Determine whether the factorization $\tau = 90 = b_1^2 \times \dim P(1, 3)$ reflects a deep structural connection between the spanning tree structure of G_H and the Poincaré group, or is a coincidence of small numbers. **Entry point:** Corollary 5.10, Section 5.

Open Problem 8.6 (New — OP-OFN-3). Derive $SU(3) \times SU(2) \times U(1)$ from the OFN decomposition $8 \oplus 3 \oplus 1 \oplus 1$ of the CP-crossing channels (Theorem 5.11). The PDL side has achieved this derivation (D46, D60, D58, D59); the OFN-side derivation remains open.

Open Problem 8.7 (New — OP-OFN-4). Derive the topological impedance α in the redshift formula $(1 + z) = \exp(\alpha d)$ from the spectral structure of Ω_{21} . Candidate: $\alpha = f(\lambda_1, \tau)$ where $\lambda_1 \approx 0.0804$ and $\tau = 90$, but f must also encode the hierarchy between the fundamental network scale and the cosmological horizon ($\alpha_{\text{phys}} \approx H_0 \cdot t_D \sim 10^{-61}$, requiring additional structure). **Epistemic status:** conjecture, not theorem.

Acknowledgements

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